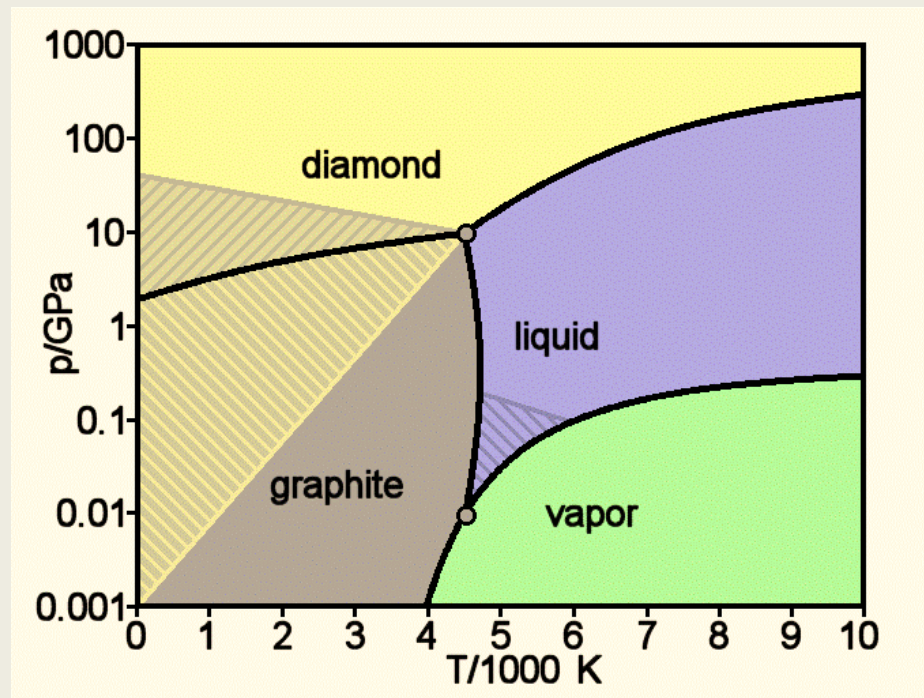


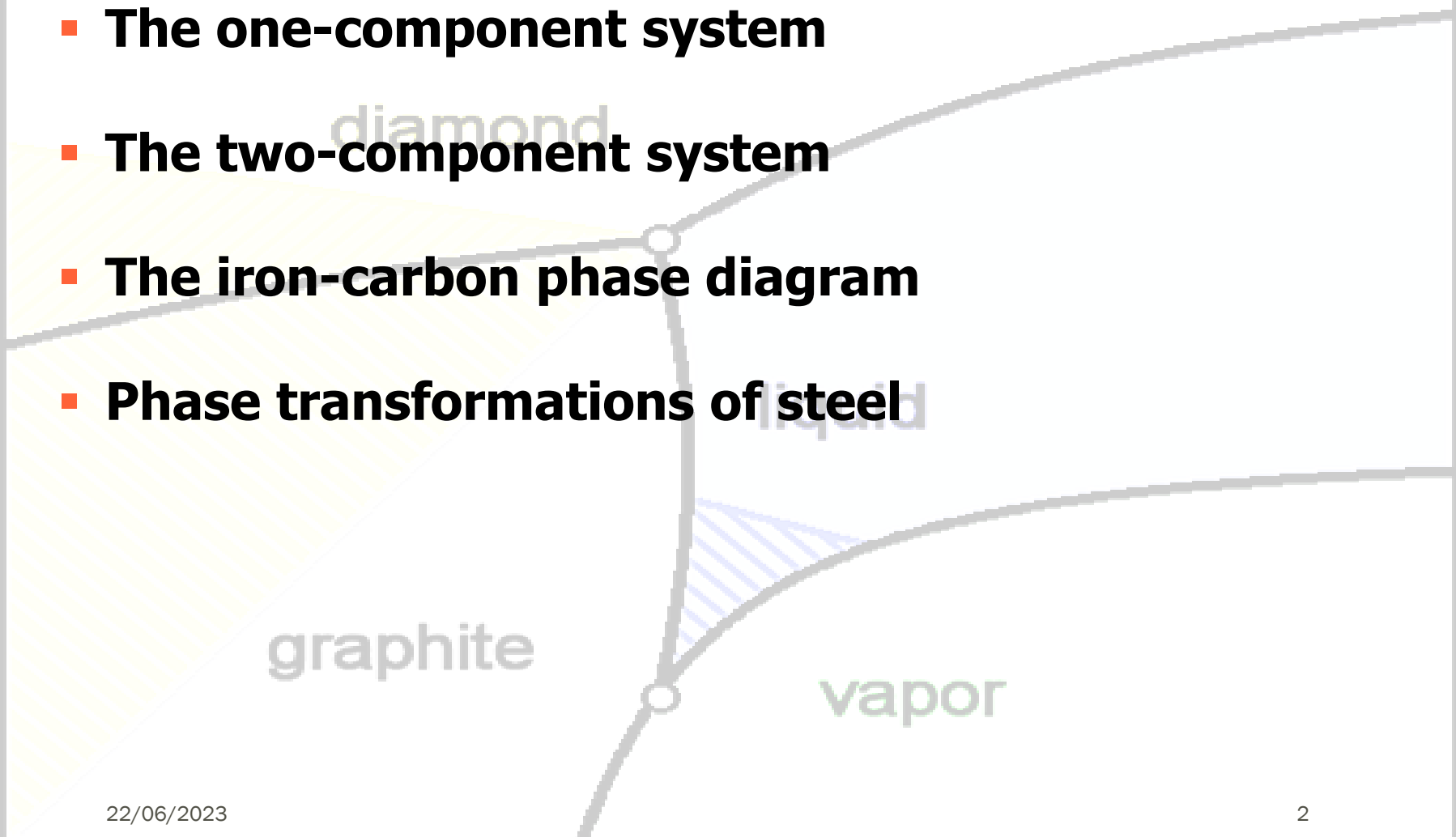
316-221 METALLURGY

Chapter 5

Phase Equilibrium Diagram



Contents

- **The one-component system**
 - **The two-component system**
 - **The iron-carbon phase diagram**
 - **Phase transformations of steel**
- 
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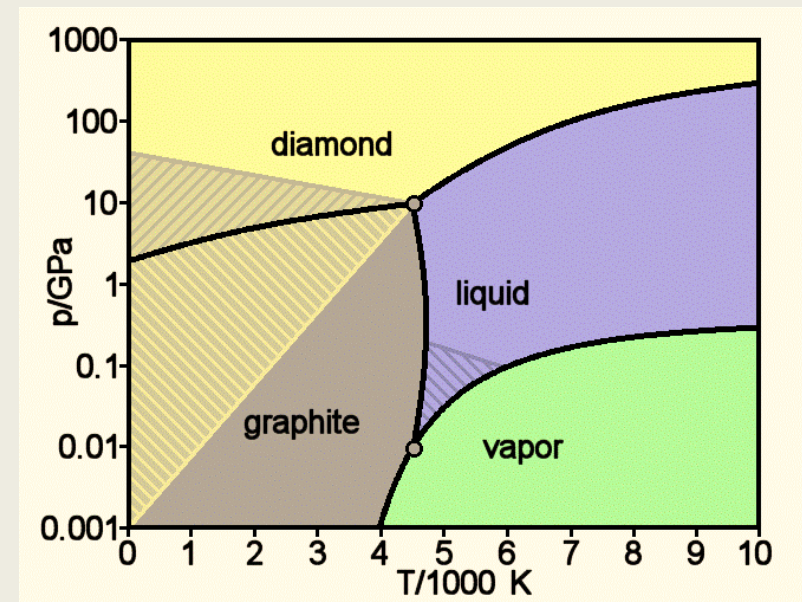
Chapter 5: Phase Equilibrium Diagram

Definitions and basic concepts

Phase equilibrium diagram is a graphic relationship between temperature and pressure or weight ratios of elements and alloys contribute to the built of the diagram

Phase diagrams provide information on the following :

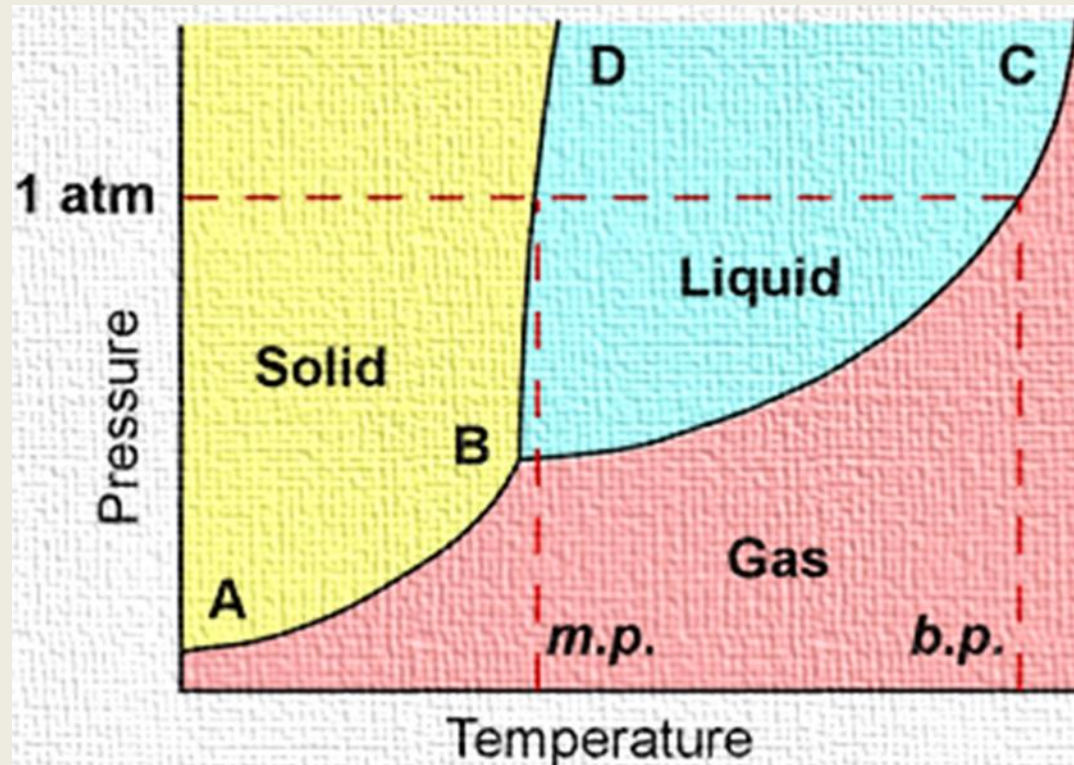
- Melting point
- Casting condition
- Crystallization condition
- Phase transformations



Chapter 5: Phase Equilibrium Diagram

Phase diagrams of the pure substance

The phases will simply be the **solid, liquid, and vapor (gas)** states of a pure substance. This is the phase diagram for a typical pure substance.



Chapter 5: Phase Equilibrium Diagram

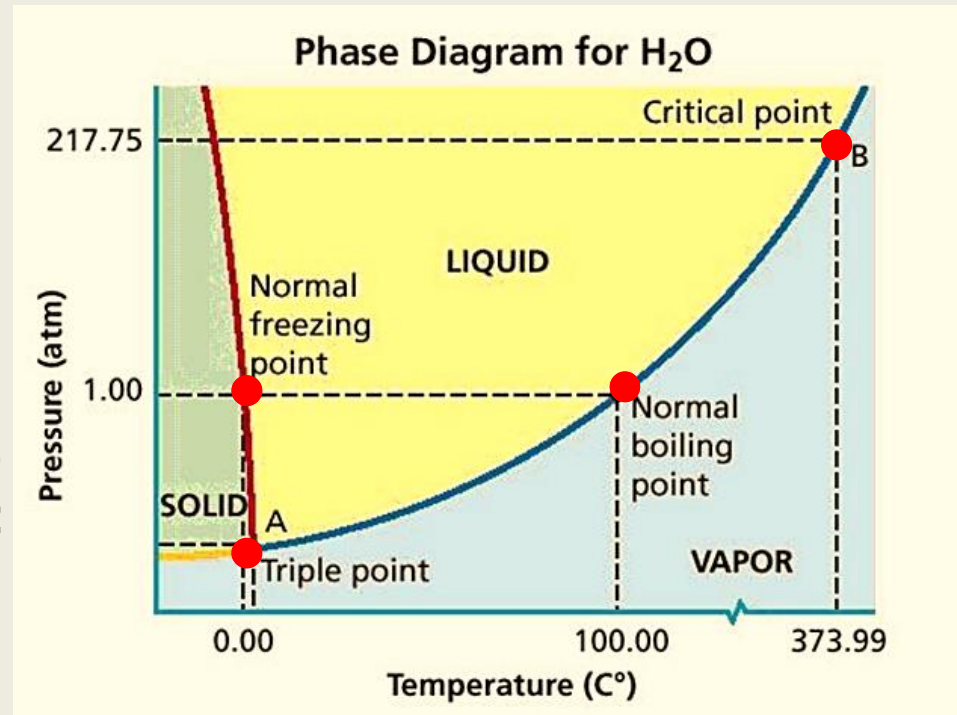
Phase diagrams of the pure substance

The critical point: at a certain pressure and temperature, it means, no separate interface between three material phases (gas, liquid, solid)

The normal freezing and boiling points: are those when the pressure is 1 atmosphere. These can be found from the phase diagram by drawing a line across at 1 atmosphere pressure

The triple point: about the three lines which meet at that point, they represent conditions of

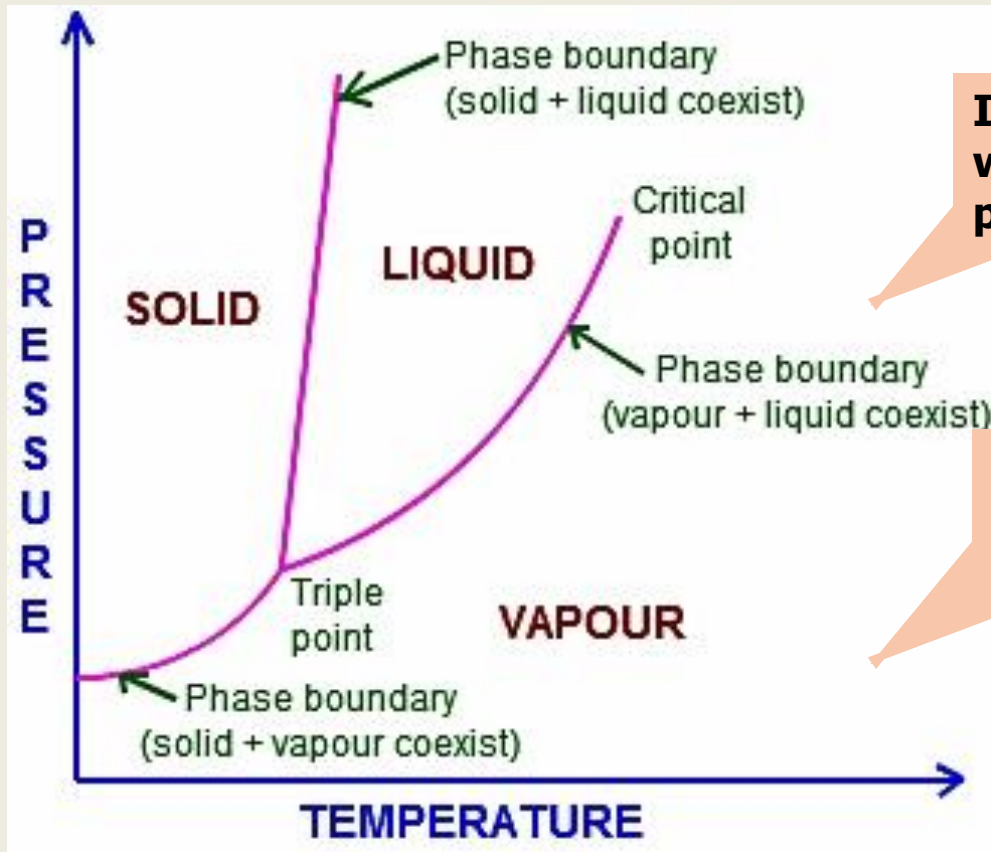
- Solid-liquid equilibrium
- Liquid-vapor equilibrium
- Solid-vapor equilibrium



Chapter 5: Phase Equilibrium Diagram

The One-Component System

In a one-component system or unary system only one chemical component is required to describe the phase relationship, for example, **water (H_2O)**, **iron (Fe)**, **methane (CH_4)**



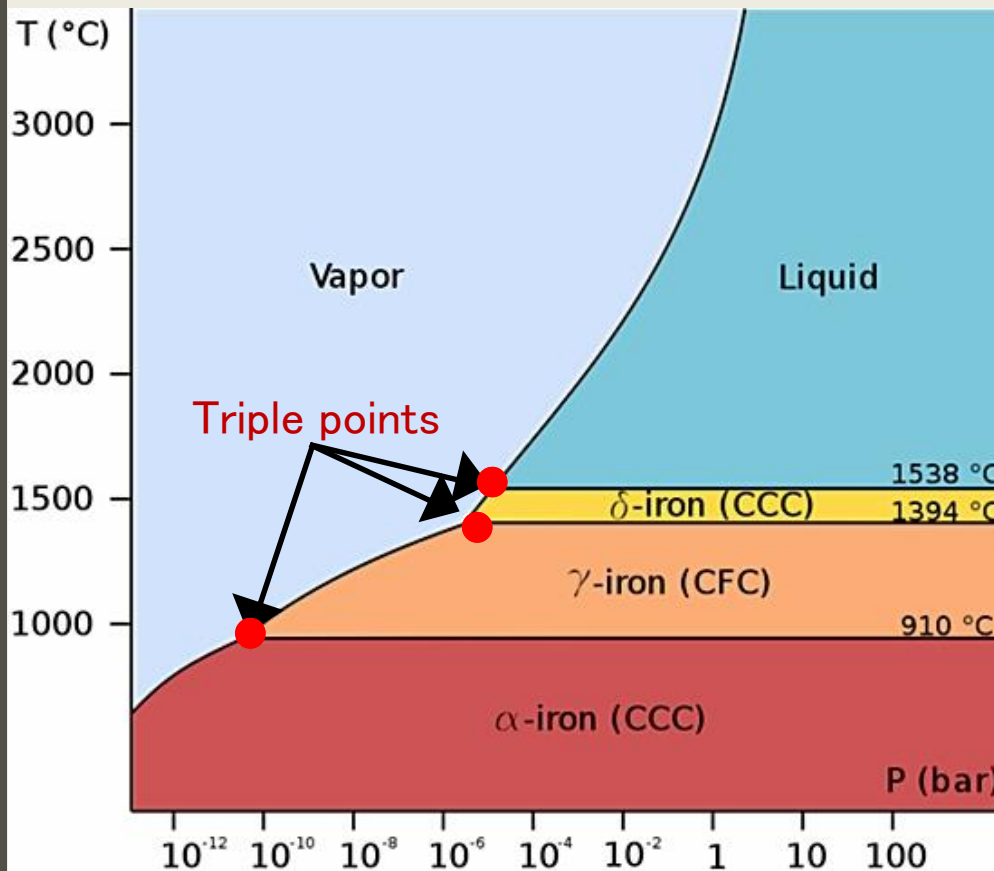
In a one-component system will not composition parameter

In a one-component system will have only two parameters (T & P)

Chapter 5: Phase Equilibrium Diagram

The One-Component System

Phase diagram of pure iron (Fe)



1538°C → (vapor+liquid+ δ -iron)

1394°C → (vapor+ δ -iron+ γ -iron)

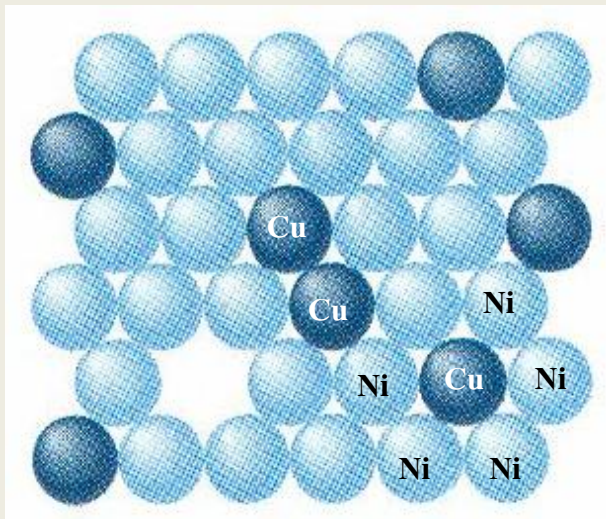
910°C → (vapor+ α -iron+ γ -iron)

Chapter 5: Phase Equilibrium Diagram

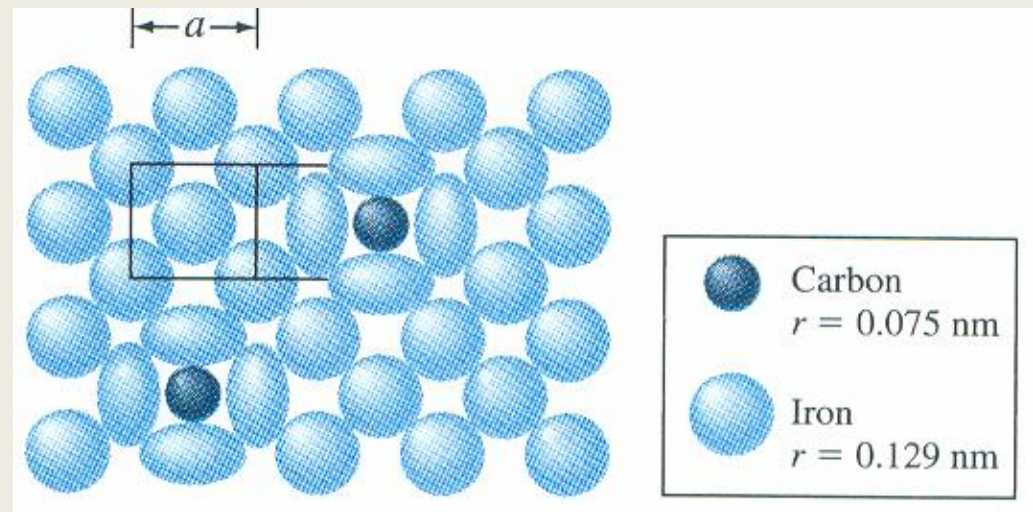
The Two-Component System

Solid solution system

Two or more atoms are homogeneous mixture and occur in solid state is call **Solid Solution**, element at more atoms is call **Solvent** and element at lower atoms is call **Solute**, **solid solution** are two types



Substitutional solid solution

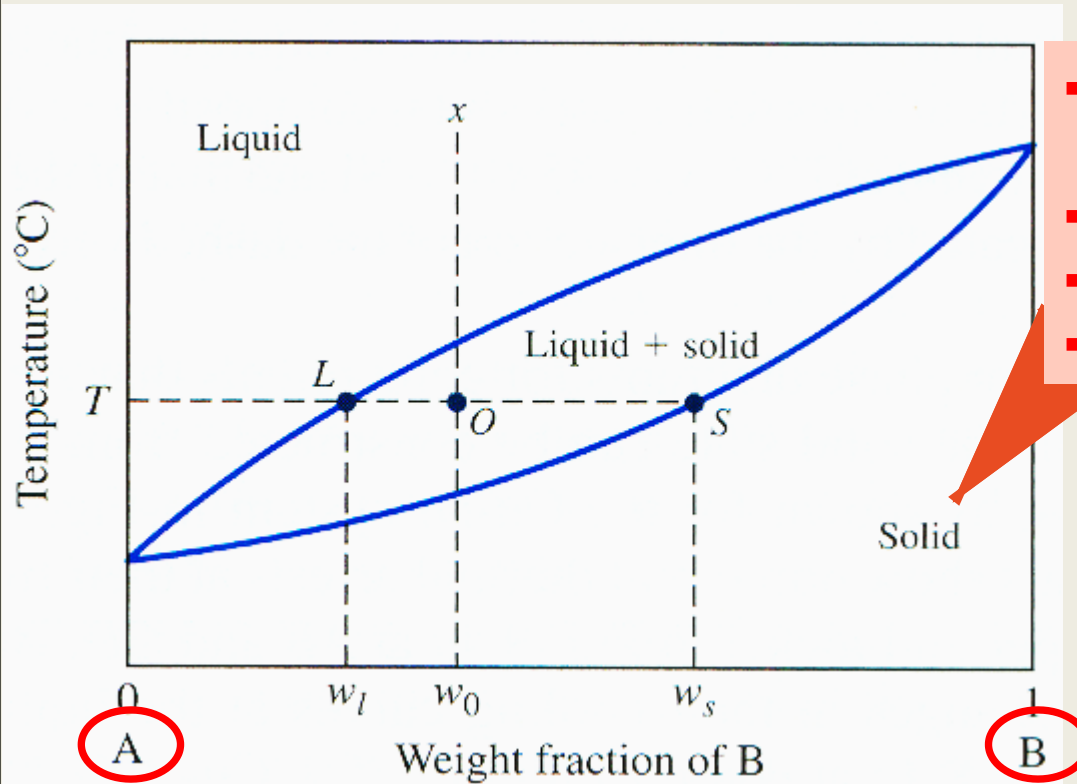


Interstitial solid solution

Chapter 5: Phase Equilibrium Diagram

The Two-Component System

Phase diagram of solid solution system

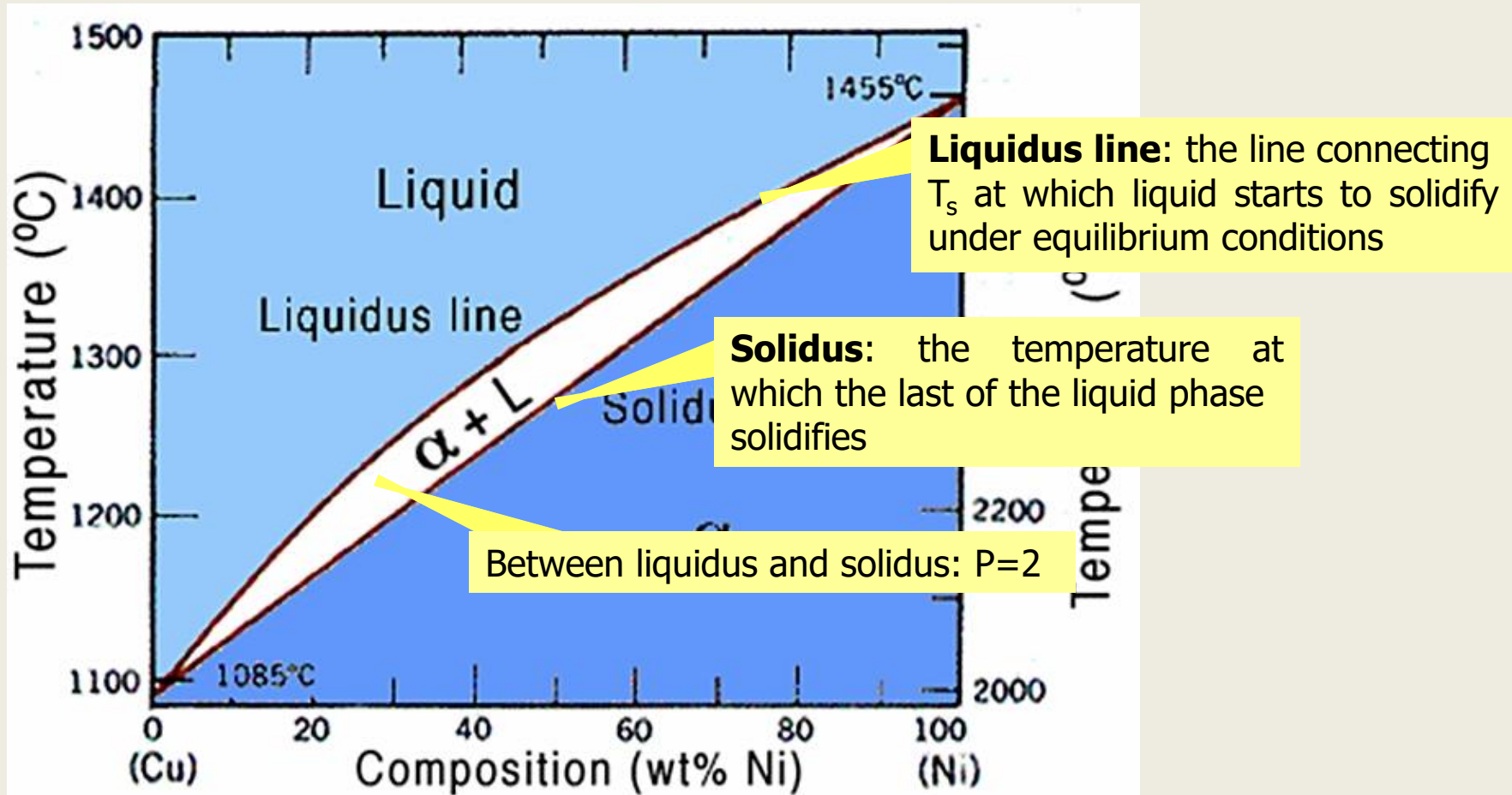


- Pure metal (**A = 100%** and **B = 100%**)
- Point **L** is Liquidus line
- Point **O** is Liquid + Solid
- Point **S** is Solidus line

Chapter 5: Phase Equilibrium Diagram

The Two-Component System

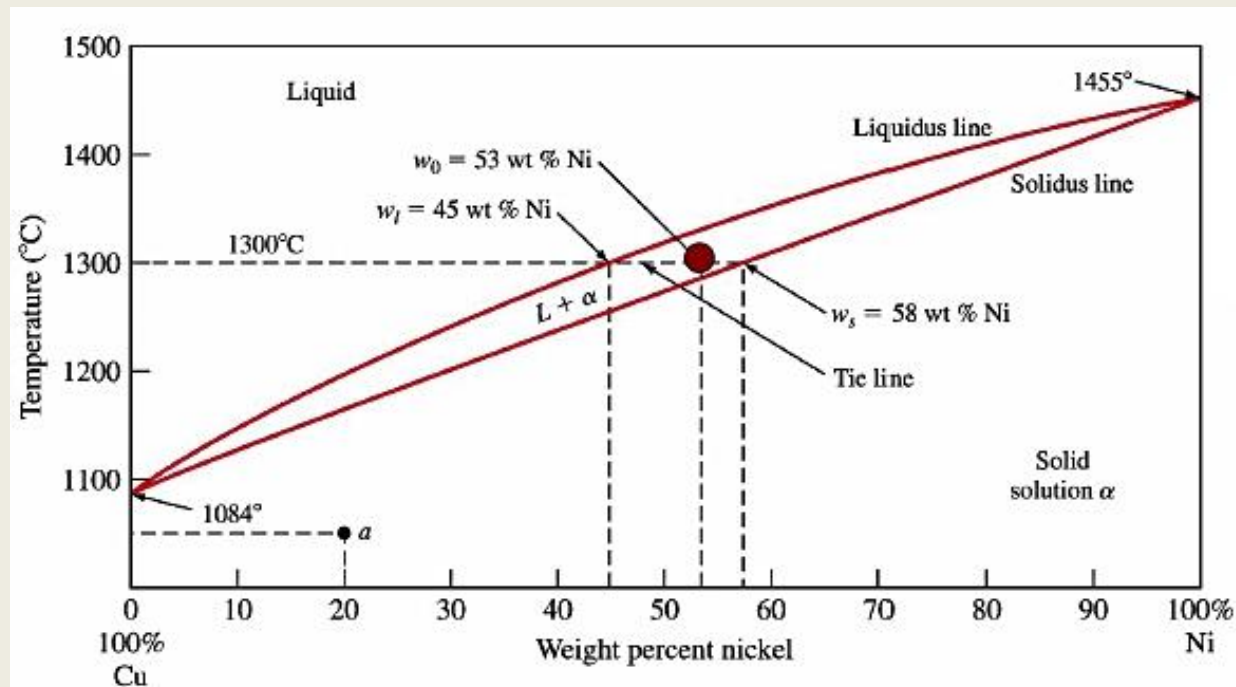
Phase diagram of solid solution system (Cu-Ni)



Chapter 5: Phase Equilibrium Diagram

The Two-Component System

- Contains both **liquid and solid** phases $\Rightarrow$ neither of these phases can have average composition **53 wt% Ni – 47 wt% Cu**
- Draw a tie line at **1300°C** $\Rightarrow$ from the graph: composition of liquid phase **$w_L = 45\%$** and solid phase **$w_S = 58\%$** at 1300°C



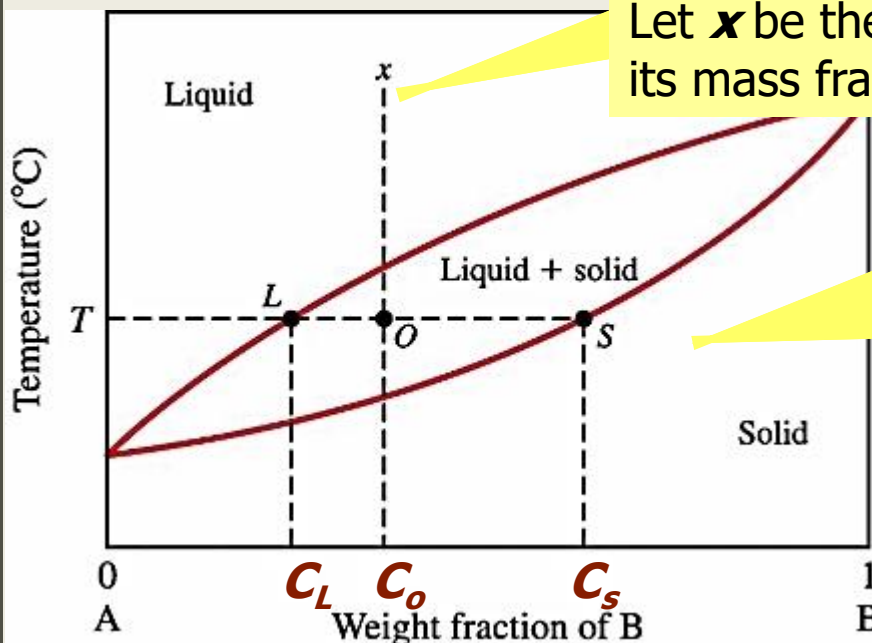
Chapter 5: Phase Equilibrium Diagram

The Two-Component System

Relative Amounts of Phases

The weight percentages of the phases in any 2 phase region can be calculated by using the **lever rule**

Consider the binary equilibrium phase diagram of elements A and B that are completely soluble in each other



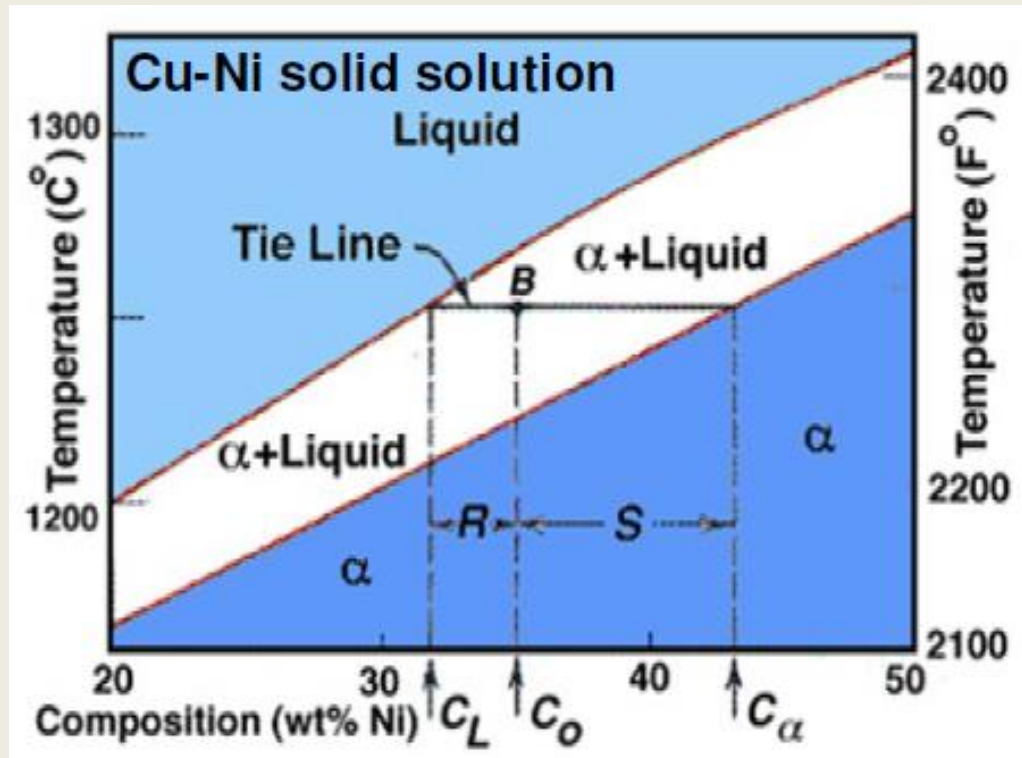
Let x be the alloy composition of interest, its mass fraction of **B** (in A) is C_0

Let T be the temperature of interest $\Rightarrow$ at T alloy x consists of a mixture of liquid (with C_L - mass fraction of **B** in liquid) and solid (C_S - mass fraction of **B** in solid phase)

Chapter 5: Phase Equilibrium Diagram

The Two-Component System

The Lever Rule



$$\% \alpha = \frac{C_0 - C_L}{C_\alpha - C_L} \times 100$$

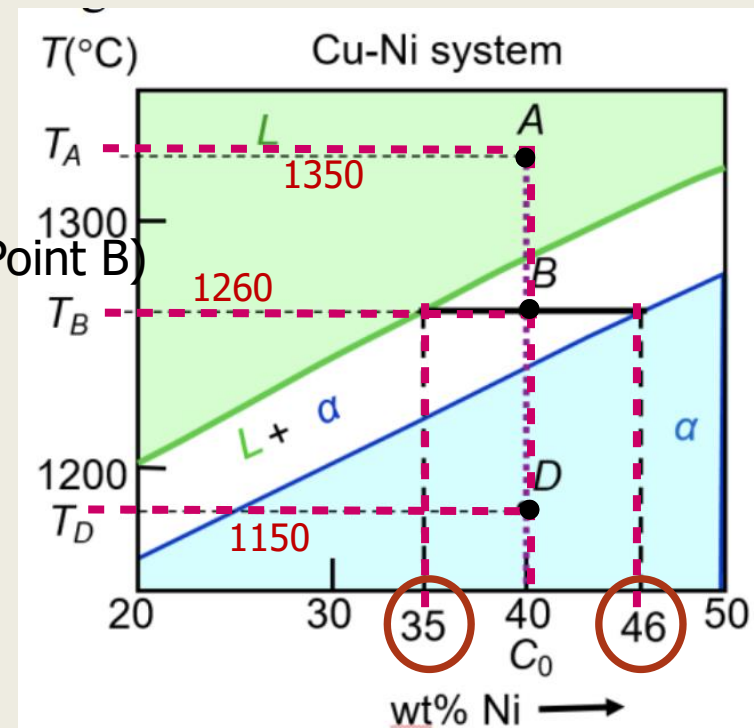
$$\% \text{liquid} = \frac{C_\alpha - C_0}{C_\alpha - C_L} \times 100$$

Chapter 5: Phase Equilibrium Diagram

The Two-Component System

Composition of phases in Cu-Ni system

- If we know T and C_0 , then we know the composition of each phase.
- At $C_0 = 40 \text{ wt\%Ni}$, $T = 1350^\circ\text{C}$
Only Liquid (L) phase (Point A)
 $C_L = C_0 = 40 \text{ wt\%Ni}$
- At $T = 1260^\circ\text{C}$, both α and L phases (Point B)
 $C_L = C_{\text{liquidus}} = 35 \text{ wt\%Ni}$
 $C_\alpha = C_{\text{solidus}} = 46 \text{ wt\%Ni}$
- At $T = 1150^\circ\text{C}$, only α phase (Point D)
 $C_\alpha = C_0 = 40 \text{ wt\%Ni}$



Chapter 5: Phase Equilibrium Diagram

The Two-Component System

Weight fraction of phases in Cu-Ni system (Lever rule)

- If we know **T** and **C_o**, then we know the **amount of each phase (wt.%)**.

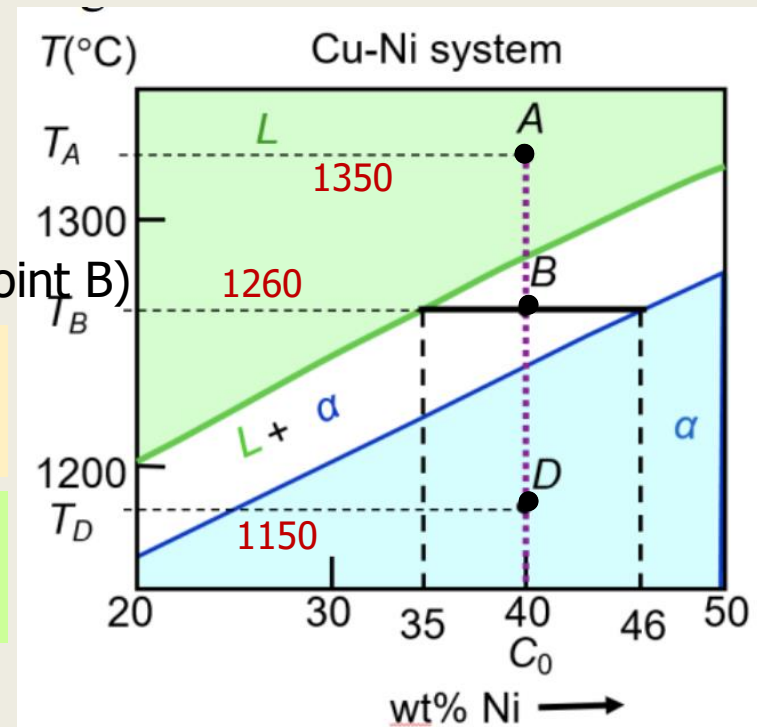
- **At C_o = 40 wt%Ni, T = 1350°C**
Only Liquid (L) phase (Point A)
W_L = 100%, W_α = 0%

- **At T = 1260°C, both L and α phases (Point B)**

$$W_L = \frac{C_\alpha - C_o}{C_\alpha - C_L} \times 100 = \frac{46 - 40}{46 - 35} \times 100 = 55\%$$

$$W_\alpha = \frac{C_o - C_L}{C_\alpha - C_L} \times 100 = \frac{40 - 35}{46 - 35} \times 100 = 45\%$$

- **At T = 1150°C, only α phase (Point D)**
W_L = 0%, W_α = 100%

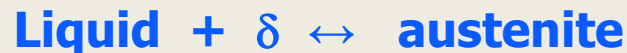


Chapter 5: Phase Equilibrium Diagram

The Iron-Carbon Phase Diagram

The diagram shows **three horizontal lines** which indicate isothermal reactions (on cooling / heating):

- First horizontal line is at **1493 °C**, where **peritectic reaction** takes place:



- Second horizontal line is at **1147 °C**, where **eutectic reaction** takes place:

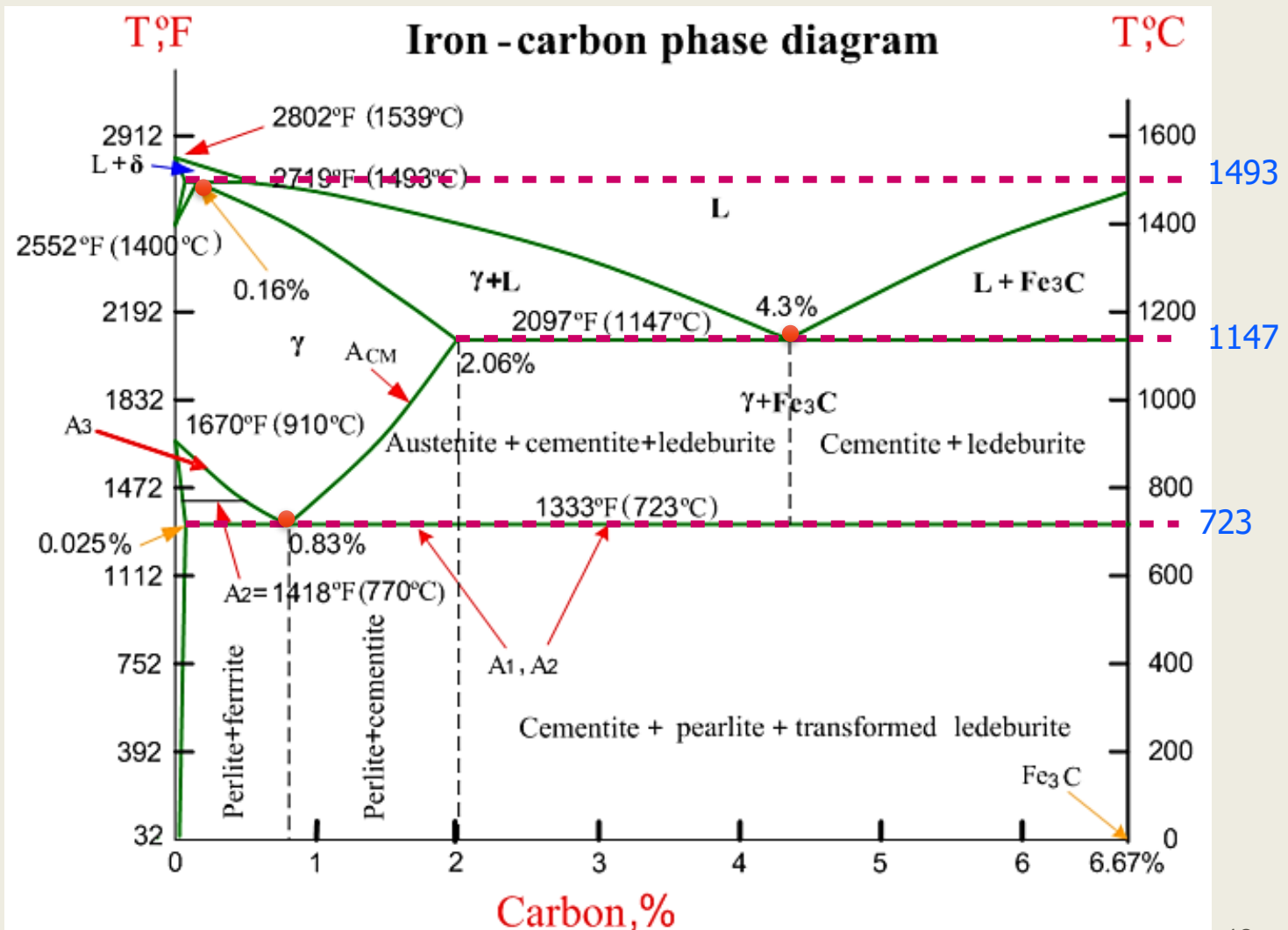


- Third horizontal line is at **723 °C**, where **eutectoid reaction** takes place:



Chapter 5: Phase Equilibrium Diagram

The Iron-Carbon Phase Diagram

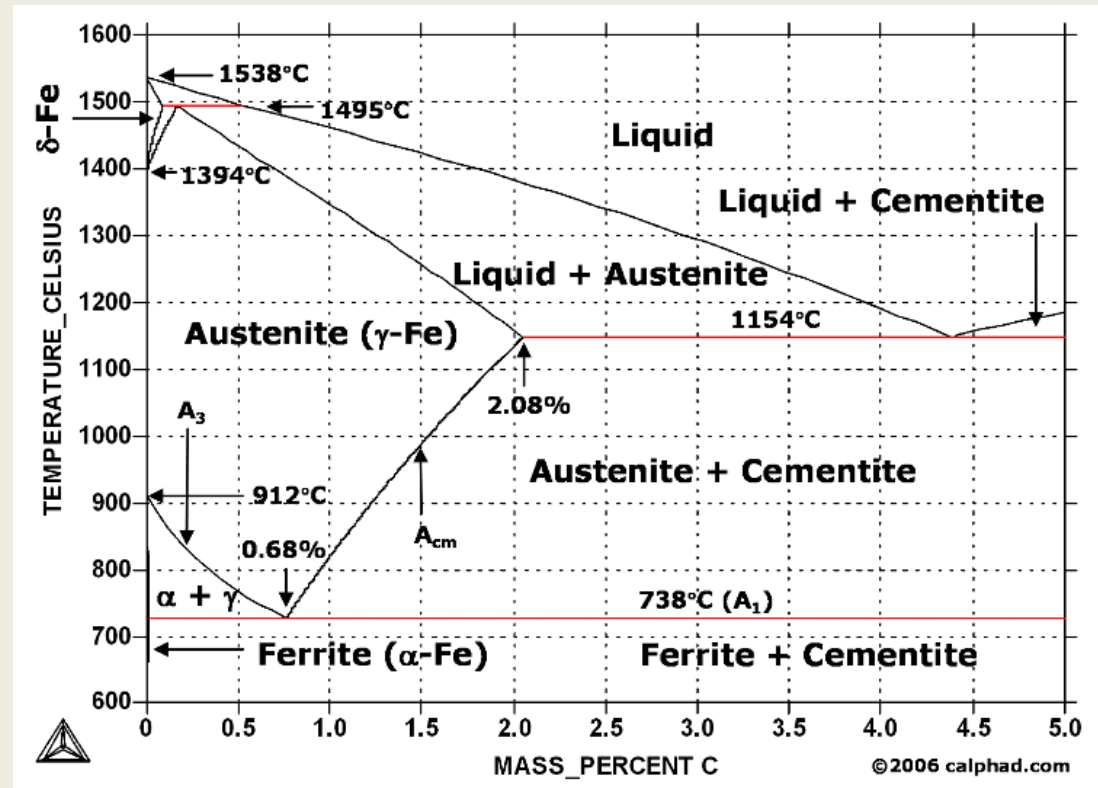


Chapter 5: Phase Equilibrium Diagram

The Iron-Carbon Phase Diagram

Various phases that appear on the **Iron-Carbon phase diagram** are as under:

- Ferrite
- Austenite
- Pearlite
- Cementite
- Martensite
- Ledeburite



Chapter 5: Phase Equilibrium Diagram

The Iron-Carbon Phase Diagram

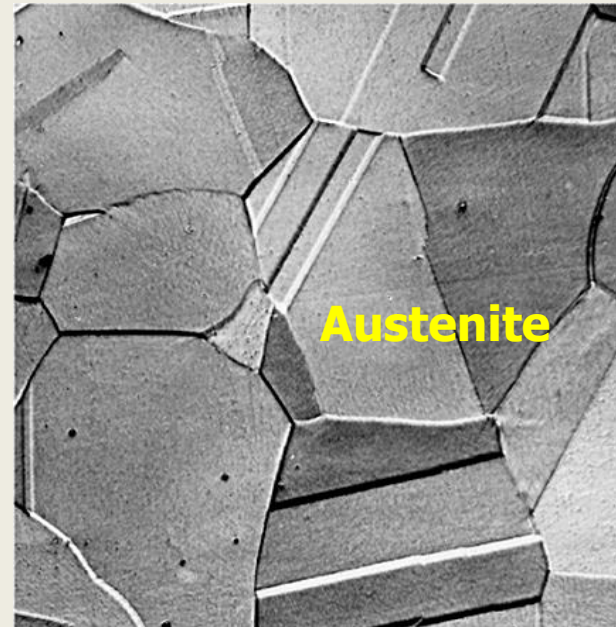
- **Ferrite** is known as α solid solution
- It is an interstitial solid solution of a small amount of carbon dissolved in α (BCC) iron
- stable form of iron below 912 °C
- The maximum solubility is 0.025 % C at 723°C and it dissolves only 0.008 %C at room temperature
- It is the softest structure that appears on the diagram
- Average properties are:
 - Tensile strength = 40,000 psi
 - Elongation = 40 %
 - Hardness = 90 HRB



Chapter 5: Phase Equilibrium Diagram

The Iron-Carbon Phase Diagram

- **Austenite** is an interstitial solid solution of **Carbon** dissolved in γ (FCC) iron
- Maximum solubility is 2.0 % C at 1130 °C
- High formability, most of heat treatments begin with this single phase
- Average properties are:
 - **Tensile strength = 150,000 psi**
 - **Elongation = 10 %**
 - **Hardness = 40 HRC**



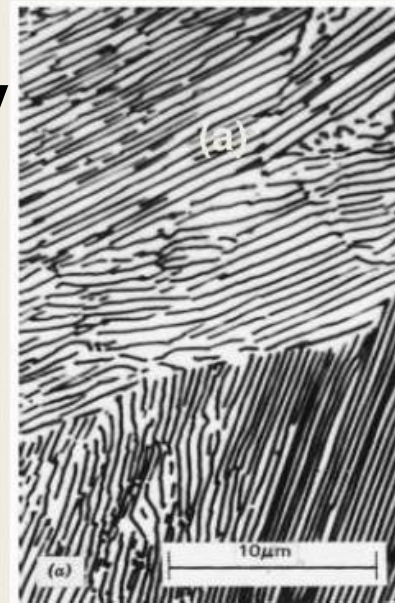
Chapter 5: Phase Equilibrium Diagram

The Iron-Carbon Phase Diagram

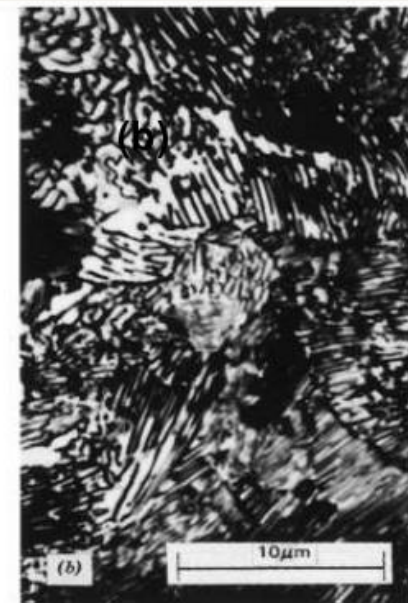
- **Pearlite** is the eutectoid mixture containing 0.76 % C and is formed at 723 °C on very slow cooling.
- It is a very fine plate like or lamellar mixture of **ferrite** and **cementite**.
- The **white ferritic** background or matrix contains thin plates of **cementite (dark)**.
- Average properties are:

- Tensile strength = 120,000 psi
- Elongation = 20 %
- Hardness = 20 HRC, 95-100 HRB

Morphology of Pearlite



(a) coarse pearlite



(b) fine pearlite 3000X

Chapter 5: Phase Equilibrium Diagram

The Iron-Carbon Phase Diagram

austenite $\leftrightarrow$ pearlite (mixture of ferrite & cementite)

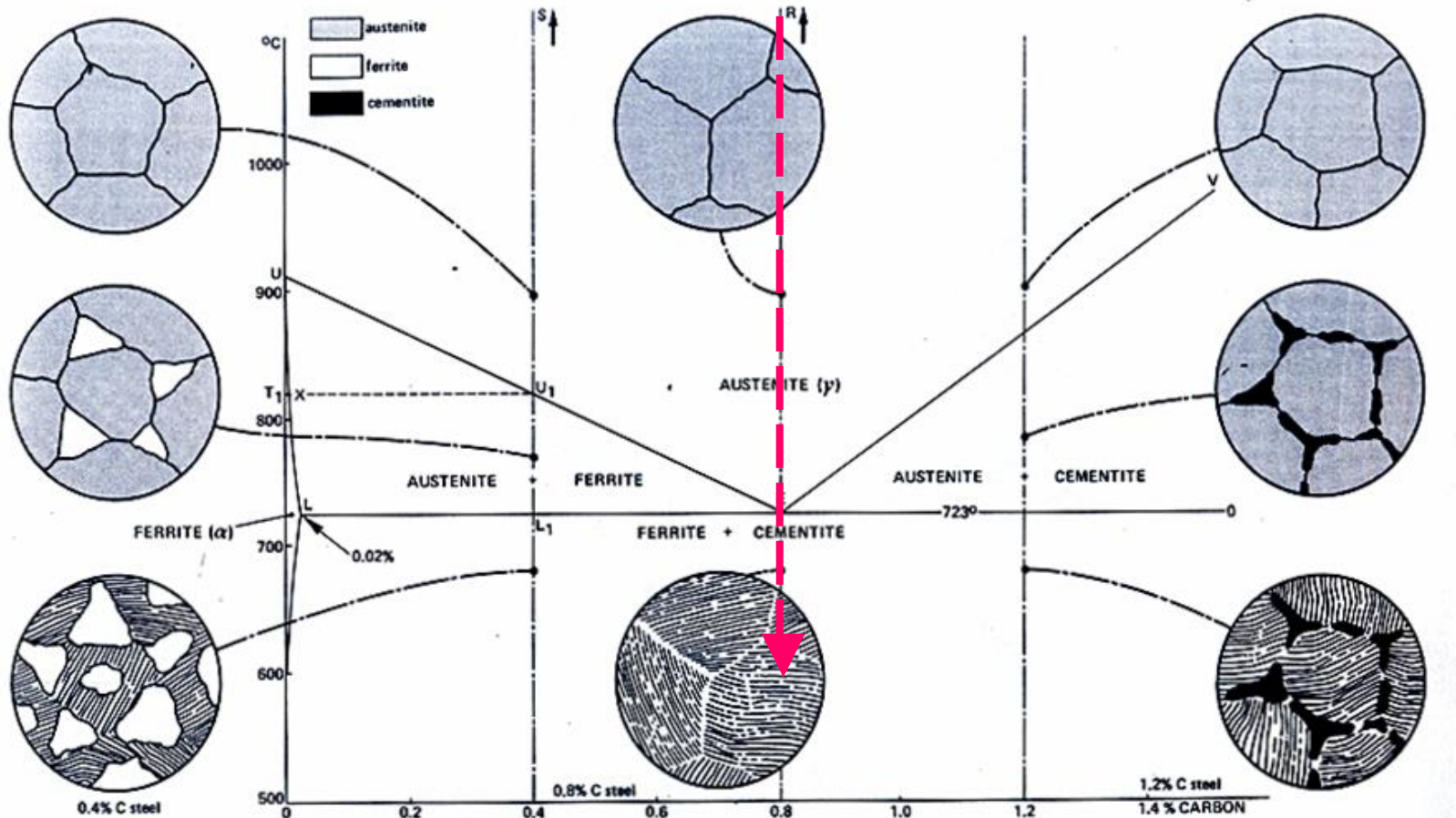


Fig. 9.3—The austenite $\rightarrow$ ferrite/cementite transformation in relation to the iron-carbon diagram.

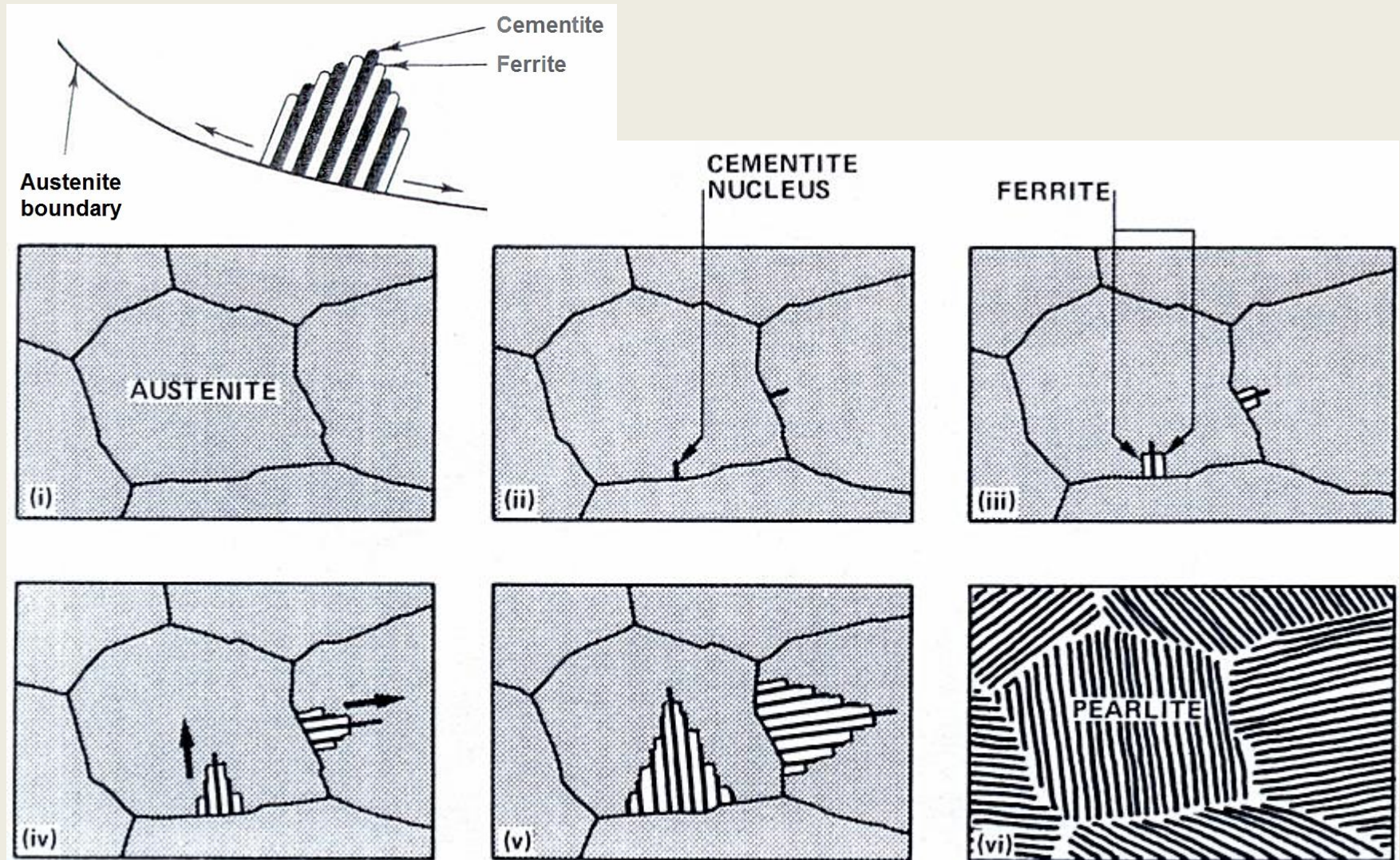
Chapter 5: Phase Equilibrium Diagram

The Iron-Carbon Phase Diagram

- In order to understand the transformation processes, consider a steel of the eutectoid composition. **0.76% carbon**, being slow cooled along **pink line**
- At the upper temperatures, only austenite is present, with the **0.76% carbon** being dissolved in solid solution within the FCC. When the steel cools through **723°C**, several changes occur simultaneously.
- The iron wants to change crystal structure from the **FCC austenite** to the **BCC ferrite**, but the ferrite can only contain **0.02% carbon** in solid solution.
- The excess carbon is rejected and forms the carbon-rich intermetallic known as **cementite**.

Chapter 5: Phase Equilibrium Diagram

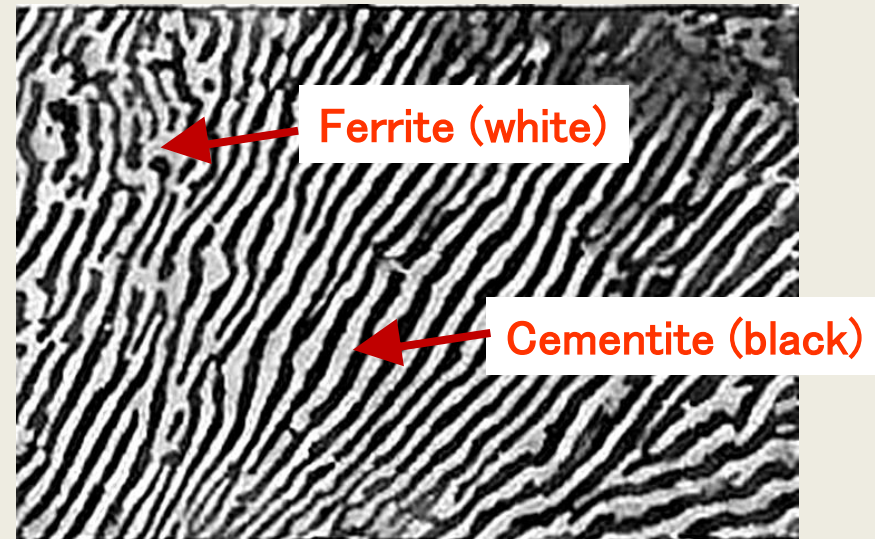
Schematic picture of the formation, nucleation and growth of pearlite



Chapter 5: Phase Equilibrium Diagram

The Iron-Carbon Phase Diagram

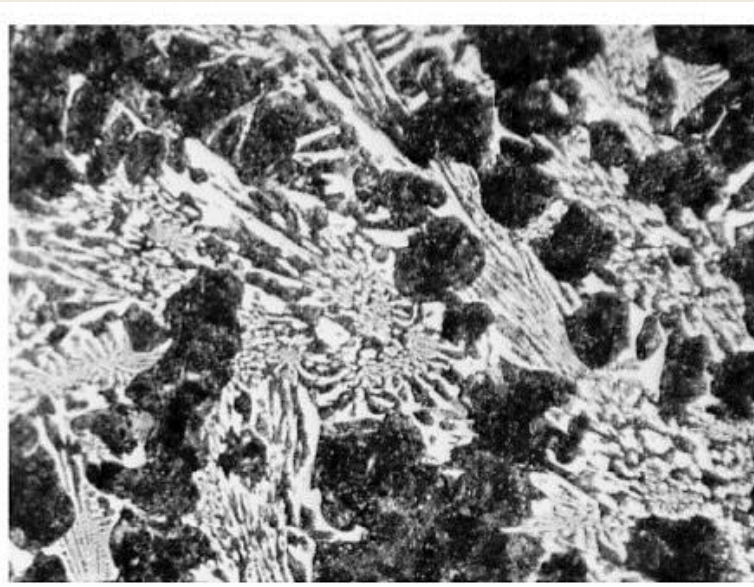
- **Cementite** or iron carbide, is **very hard, brittle** intermetallic compound of iron & carbon, as Fe_3C , contains **6.67 % C**.
- It is the **hardest structure** that appears on the diagram.
- ❑ Its crystal structure is **orthorhombic**.
- ❑ It is has
 - low tensile strength
(**approx. 5,000 psi**)
 - high compressive strength



Chapter 5: Phase Equilibrium Diagram

The Iron-Carbon Phase Diagram

- **Ledeburite** is a mixture of 4.3% carbon in iron and is a eutectic mixture of **austenite** and **cementite**.
- It contains 4.3wt.%C and is formed at 1147 °C.

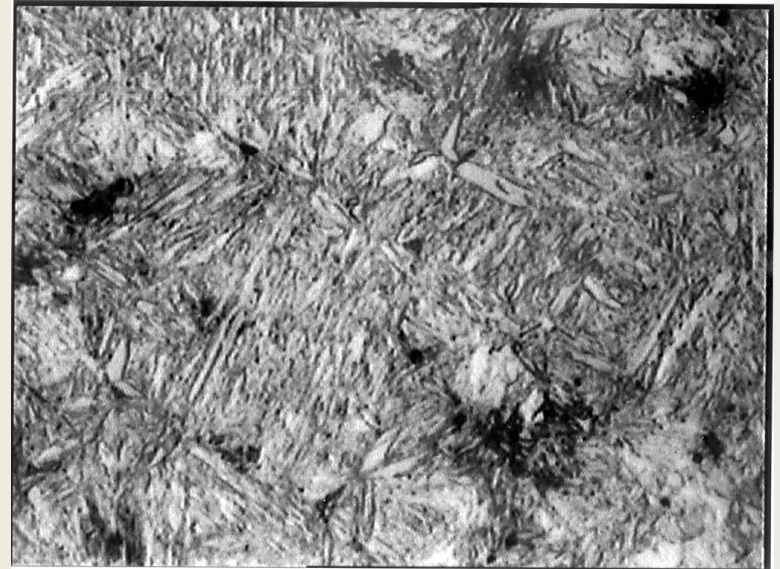


- It is the eutectic lamellar mixture of austenite (light phase) and cementite (dark phase)
- It is unstable below 727°C and transforms into **α -ferrite and cementite.**

Chapter 5: Phase Equilibrium Diagram

The Iron-Carbon Phase Diagram

- ❑ **Martensite** - a supersaturated solid solution of **carbon in ferrite**.
- ❑ It is formed when steel is cooled so rapidly that the change **from austenite to pearlite** is suppressed.
- ❑ The interstitial carbon atoms distort **the BCC ferrite into a BC- tetragonal structure (BCT)**.; responsible for the hardness of quenched steel.

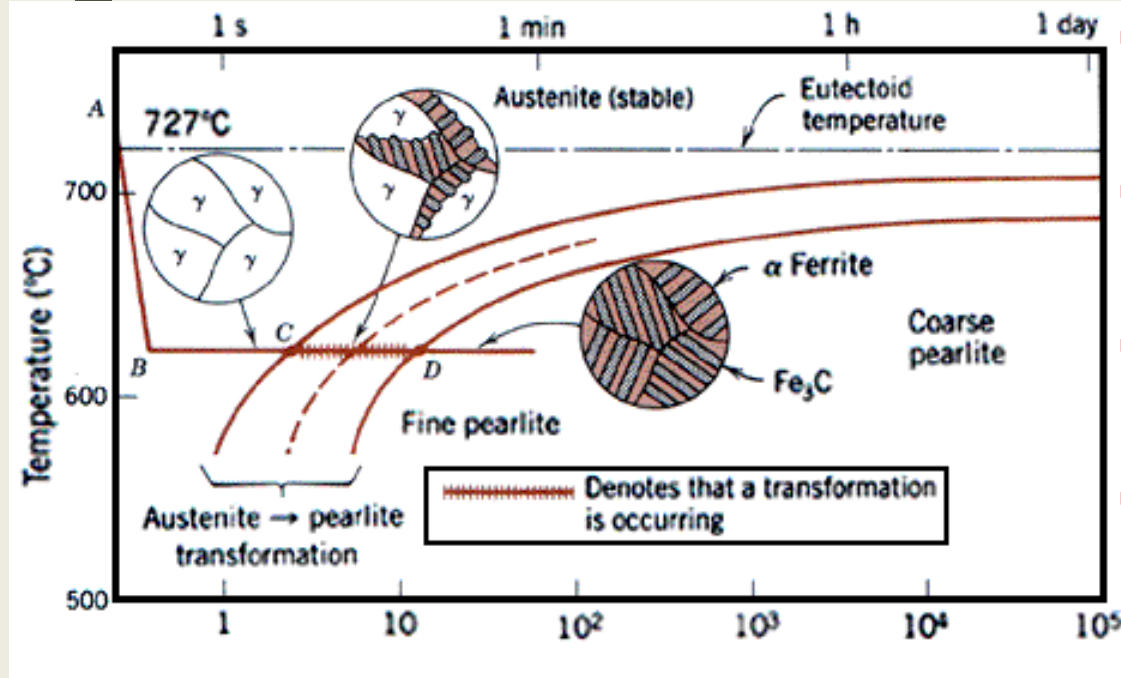


0.35%C Steel, water-quenched from 870 °C

Chapter 5: Phase Equilibrium Diagram

Iron-Iron Carbide Phase Transformations on the TTT Diagram

The time-temperature transformation curves (**TTT Diagram**) correspond to the start and finish of transformations which extend into the range of temperatures where **austenite** transforms to **pearlite**

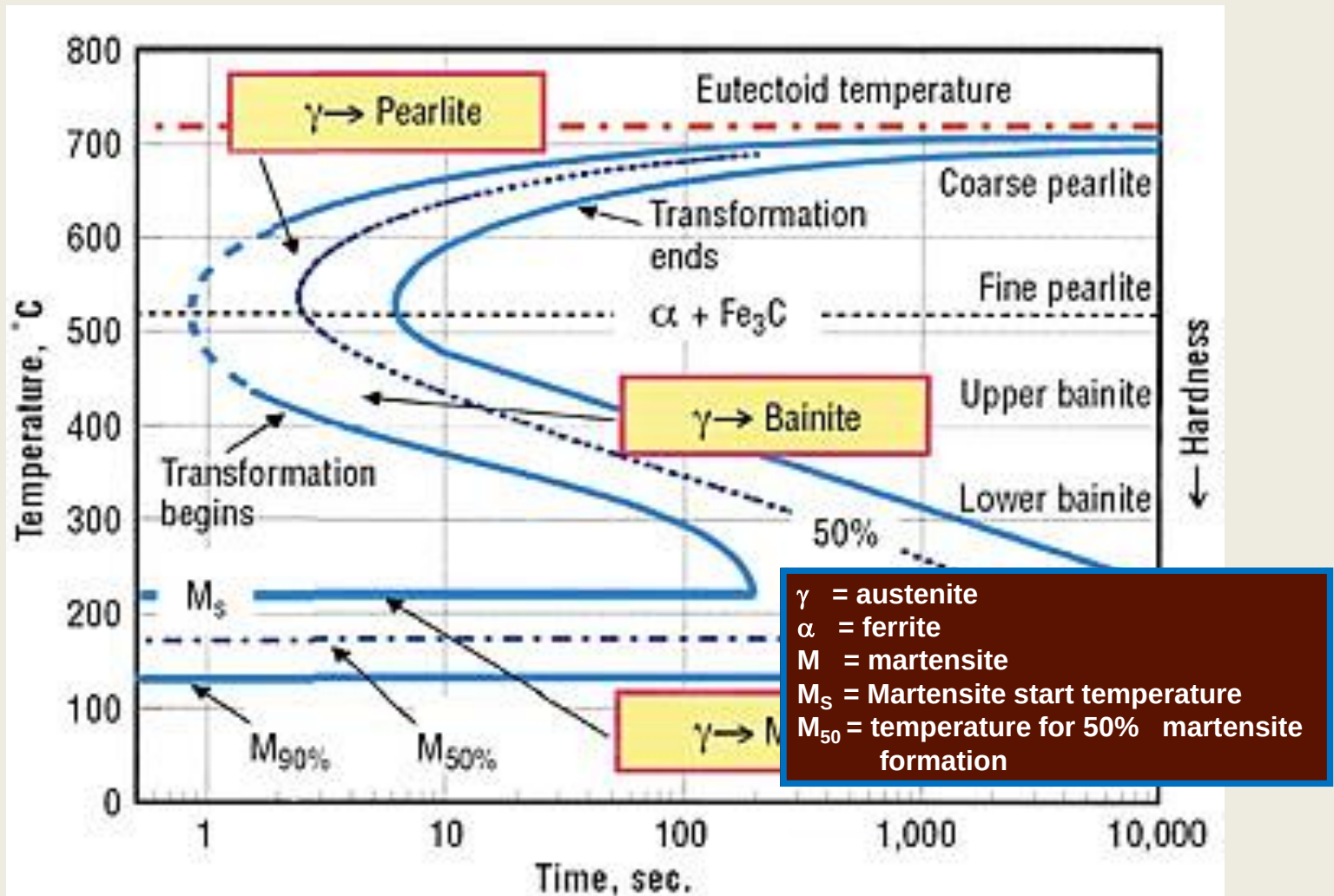


- Above 550 °C, **austenite** transforms completely to **pearlite**
- Below 550 °C, both **pearlite** and **bainite** are formed
- below 450 C, only **bainite** is formed
- The horizontal line **C-D** that runs between the two curves marks the beginning and end of isothermal transformations

Fe-Fe₃C TTT Diagram

Chapter 5: Phase Equilibrium Diagram

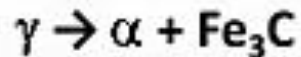
Iron-Iron Carbide Phase Transformations on the TTT Diagram



Chapter 5: Phase Equilibrium Diagram

Transformations of austenite to pearlite

Transformations of austenite :

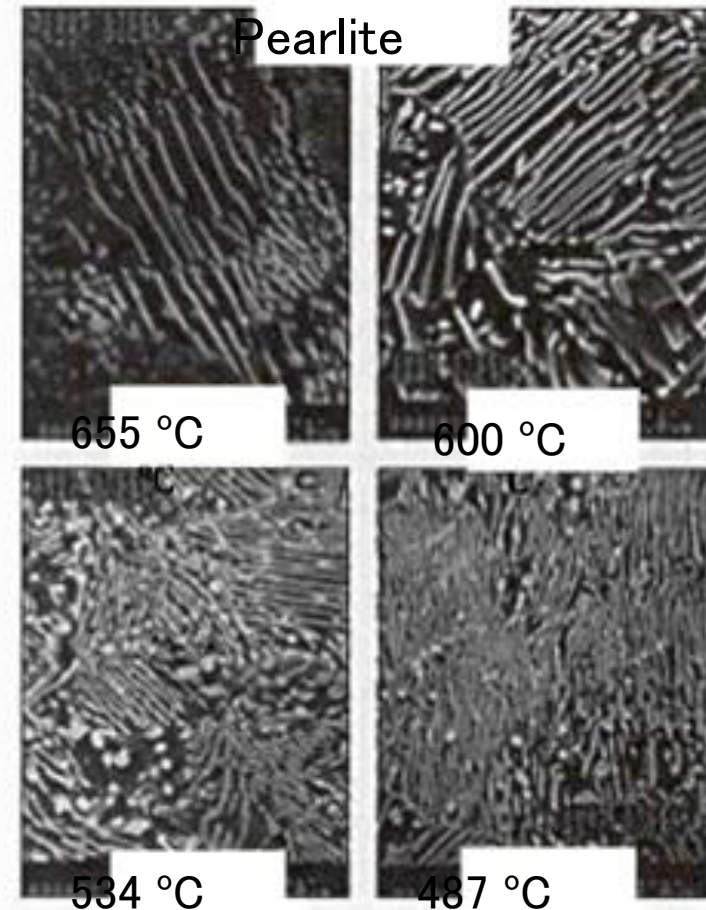


■ At slightly lower T below 727 °C : $\Delta T \ll$

- Coarse pearlite
: nucleation rate is very low.
: diffusion rate is very high.

■ As the T (trans. temp.) decreases to 500 °C

- Fine pearlite
: nucleation rate increases.
: diffusion rate decreases.



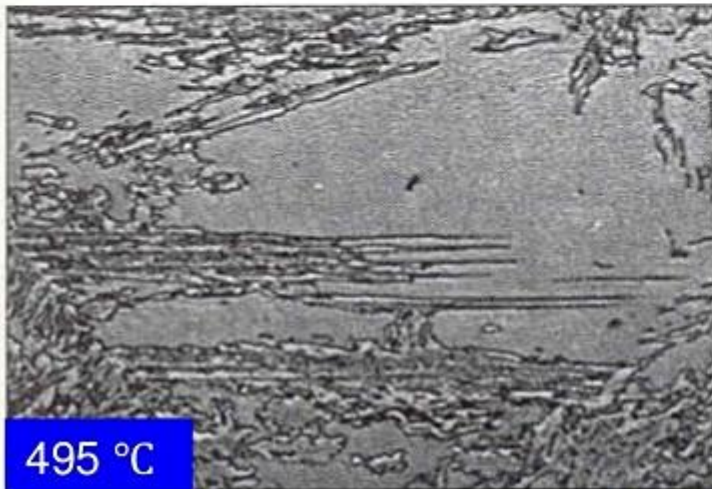
Chapter 5: Phase Equilibrium Diagram

Transformations of austenite to bainite

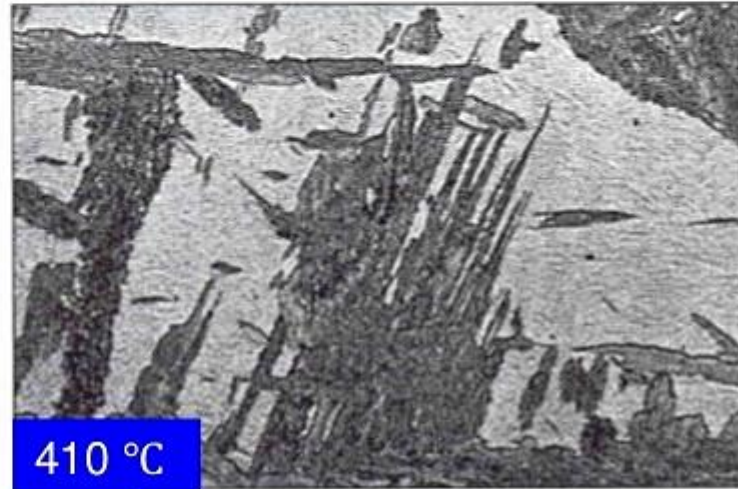
■ At further lower temperatures, $250\text{ }^{\circ}\text{C} < T_t < 500\text{ }^{\circ}\text{C}$, below the nose in TTT diagram.

- Driving force for the transformation ($\gamma \rightarrow \alpha + \text{Fe}_3\text{C}$) is very high.
- Diffusion rate is very low.
- Nucleation rate is very high.

$\gamma \rightarrow \alpha + \text{Fe}_3\text{C}$ (But not in the form of alternate layers)
: Bainite ; cementite in the form of needle type.



495 °C



410 °C

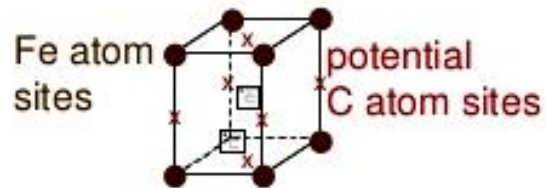
bainite

Chapter 5: Phase Equilibrium Diagram

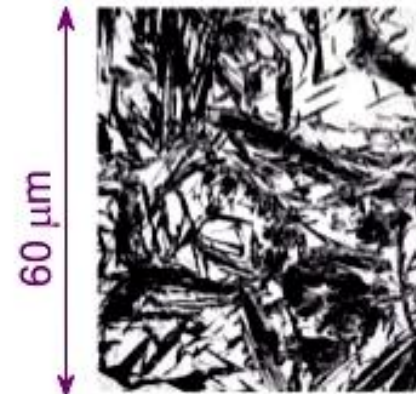
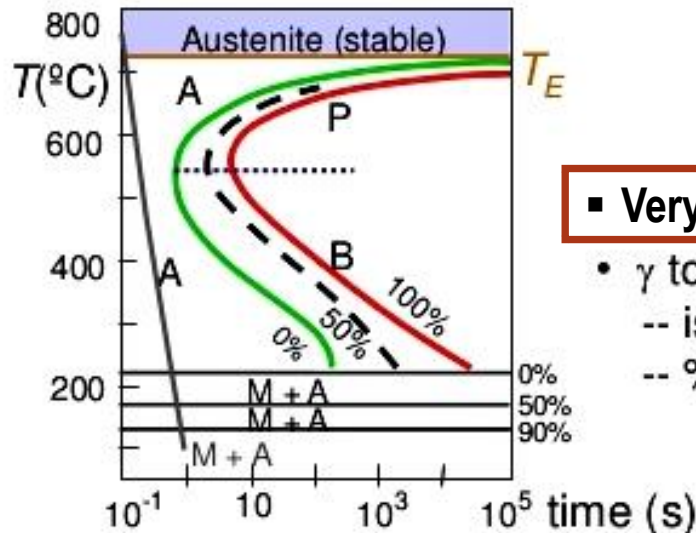
Transformations of austenite to martensite

When the austenite is quenched to temperature below M_s

- **Martensite:**
-- γ (FCC) to Martensite (BCT)



- Isothermal Transf. Diagram



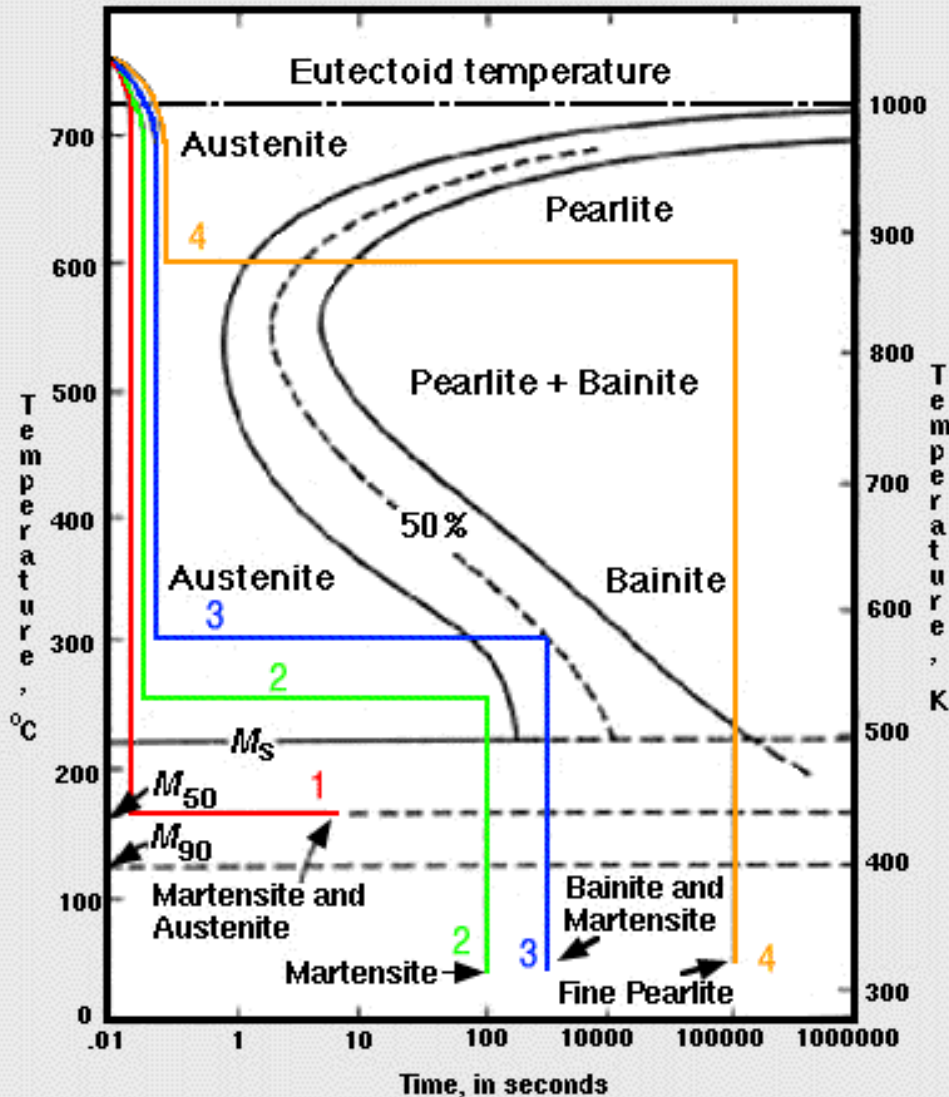
— Martensite needles
— Austenite

▪ Very hard and brittle phase

- γ to martensite (M) transformation..
-- is rapid! (diffusionless)
-- % transf. depends only on T to which rapidly cooled

Chapter 5: Phase Equilibrium Diagram

Iron-Iron Carbide Phase Transformations on the TTT Diagram



Example

What transformations happen in:

- Path 1 (Red line)
- Path 2 (Green line)
- Path 3 (Blue line)
- Path 4 (Orange line)

a. The specimen is cooled rapidly to (433K) **160°C**. Since 433 K is the temperature at which half of the **austenite transforms to martensitic**

b. The specimen is held at 523 K (**250 °C**) which is not long enough to form bainite. Therefore, the second quench from 523 K to room temp. develops a **martensitic structure**

c. An isothermal hold at 573 K (**300°C**) produces a **half-bainite and half-austenite** structure, cooling quickly would result in a final structure of **martensite and bainite**

d. Austenite converts completely to **fine pearlite** at 873 K (**600°C**), this phase is stable and will not be changed at 873 K. The final structure, when cooled, is **fine pearlite**